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Monte Carlo Investigation of Magnetocaloric Properties and Critical Behavior in Ni-Doped GdMnO₃ Perovskites: A Comparative Study with Experimental Data --Manuscript Draft--

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Abstract:	This research work deals with the systematic study of the magnetocaloric effect and criticality of the Ni-doped gadolinium manganite, GdMnO₃ doped with Nickel, GdMn_{1-x}Ni_xO₃ where x equals 0.0, 0.05, and 0.10, using classical Monte Carlo simulations based on three-dimensional Heisenberg model. The simulations have been carried out using the A-type antiferromagnetic structure with nearest-neighbor ferromagnetic (J_1) and next-nearest-neighbor antiferromagnetic (J_2) exchange interactions whose values were taken from the experimental transition temperature. The calculation of the staggered magnetization, magnetic susceptibility, and Binder cumulant was done as a function of temperature at various doping levels. The magnetic entropy change, adiabatic temperature change, relative cooling power, and refrigerant capacity were obtained from the magnetic moment calculations. Our results reveal that Ni doping significantly modifies the magnetic properties: the transition temperature (TN) decreases from 51.8 K (x=0.0) to 44.2 K (x=0.05) before increasing to 49.1 K (x=0.10), in excellent agreement with experimental values (52 K, 44 K, and 49 K, respectively). The maximum entropy change at 10 T field reaches 1.23 J kg⁻¹ K⁻¹ for x=0.0, 0.98 J kg⁻¹ K⁻¹ for x=0.05, and 1.15 J kg⁻¹ K⁻¹ for x=0.10. The critical exponents (β, γ, δ) were determined using power-law fitting and scaling analysis, yielding values consistent with the 3D Ising universality class. The calculated critical exponents $\beta = 0.325$-0.338, $\gamma = 1.237$-1.310, and $\delta = 4.78$-5.02 confirm the short-range magnetic interactions in these systems. The RCP values for 10 T fields are 45.7, 38.2, and 42.5 J kg⁻¹ for x = 0.0, 0.05, and 0.10, respectively, signifying a good magnetocaloric effect. The reason behind the non-monotonic behavior of magnetic properties due to Ni-doping can be explained by the fact that at lower doping values there exists an increase in the ferromagnetic behavior of the material (Mn³⁺-O-Ni³⁺ double exchange), while for higher Ni doping there is antiferromagnetic superexchange interaction (Ni³⁺-O-Ni³⁺). This detailed MC simulation study has provided an insight into the magnetic behavior and magnetocaloric effects in Ni-doped GdMnO₃.

Monte Carlo Investigation of Magnetocaloric Properties and Critical Behavior in Ni-Doped GdMnO₃ Perovskites: A Comparative Study with Experimental Data

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Abstract

This research work deals with the systematic study of the magnetocaloric effect and criticality of the Ni-doped gadolinium manganite, GdMnO₃ doped with Nickel, GdMn_{1-x}Ni_xO₃ where x equals 0.0, 0.05, and 0.10, using classical Monte Carlo simulations based on three-dimensional Heisenberg model. The simulations have been carried out using the A-type antiferromagnetic structure with nearest-neighbor ferromagnetic (J_1) and next-nearest-neighbor antiferromagnetic (J_2) exchange interactions whose values were taken from the experimental transition temperature. The calculation of the staggered magnetization, magnetic susceptibility, and Binder cumulant was done as a function of temperature at various doping levels. The magnetic entropy change, adiabatic temperature change, relative cooling power, and refrigerant capacity were obtained from the magnetic moment calculations. Our results reveal that Ni doping significantly modifies the magnetic properties: the transition temperature (T_N) decreases from 51.8 K ($x=0.0$) to 44.2 K ($x=0.05$) before increasing to 49.1 K ($x=0.10$), in excellent agreement with experimental values (52 K, 44 K, and 49 K, respectively). The maximum entropy change at 10 T field reaches 1.23 J kg⁻¹ K⁻¹ for $x=0.0$, 0.98 J kg⁻¹ K⁻¹ for $x=0.05$, and 1.15 J kg⁻¹ K⁻¹ for $x=0.10$. The critical exponents (β , γ , δ) were determined using power-law fitting and scaling analysis, yielding values consistent with the 3D Ising universality class. The calculated critical exponents $\beta = 0.325-0.338$, $\gamma = 1.237-1.310$, and $\delta = 4.78-5.02$ confirm the short-range magnetic interactions in these systems. The RCP values for 10 T fields are 45.7, 38.2, and 42.5 J kg⁻¹ for $x = 0.0$, 0.05, and 0.10, respectively,

signifying a good magnetocaloric effect. The reason behind the non-monotonic behavior of magnetic properties due to Ni-doping can be explained by the fact that at lower doping values there exists an increase in the ferromagnetic behavior of the material ($\text{Mn}^{3+}\text{-O-Ni}^{3+}$ double exchange), while for higher Ni doping there is antiferromagnetic superexchange interaction ($\text{Ni}^{3+}\text{-O-Ni}^{3+}$). This detailed MC simulation study has provided an insight into the magnetic behavior and magnetocaloric effects in Ni-doped GdMnO_3 .

Keywords: Monte Carlo simulation; Magnetocaloric effect; Critical exponents; Magnetic refrigeration; A-type antiferromagnet

1. Introduction

Efforts have been focused on finding efficient magnetocaloric materials for cryogenic magnetic cooling systems owing to the rise in environmental-friendly technology [1,2]. Different kinds of materials have been investigated; among these, RMnO_3 , a type of rare-earth manganites with an orthorhombic perovskite structure has gained prominence for its magnetism and magnetoelectric characteristics [3-5]. The particular interest lies in GdMnO_3 (GMO), which has a complicated magnetic phase transition involving multiple phases of magnetic transitions from PM state to ICAFM state at $T_N \approx 42$ K, then further to c-AFM state at $T_c \approx 23$ K [6-8]. The magnetic structure is defined by the competition between nearest neighbor ferromagnetic (NN-FM) and next nearest neighbor antiferromagnetic (NNN-AFM) interactions within the MnO_2 layers, giving rise to the A-type antiferromagnetic order [9,10].

Recently performed experiments showed that doping of the Mn site could greatly influence the magnetic properties of GdMnO_3 [11,12]. Doping with Ni^{3+} ions (radius $0.56 \text{ \AA}$) at the Mn^{3+} ions site (radius $0.64 \text{ \AA}$) results in local distortions of the lattice and change in the exchange interactions, which leads to controllable magnetic properties [13]. It was shown experimentally by Solanki et al. [14] that doping of $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ ($x = 0.0, 0.05, 0.10$) results in the changes in structural parameters, electrical resistivity, and magnetic ordering temperature. However, there is not yet a complete picture of magnetic interactions and magnetocaloric effects on the microscopic scale. This kind of information can be obtained by means of Monte Carlo (MC) simulation techniques [15,16], through using the classical version of the Heisenberg model and considering the exchange interactions, the MC approach will be able to simulate experimental data as well as predict new properties [17]. The magnetocaloric effect (MCE), which includes magnetic entropy change (ΔS)

and adiabatic temperature change (ΔT_{ad}), is particularly important in assessing the cooling properties of magnetic materials [18,19]. Here, we report a detailed Monte Carlo simulation on the magnetic and magnetocaloric properties of GdMnO_3 doped with Ni. The aim here is to: (i) deduce the exchange coupling parameters based on the transition temperatures of the experiment, (ii) obtain the magnetization and susceptibility of the compound as a function of temperature, (iii) compute the magnetocaloric properties such as ΔS , ΔT_{ad} , RCP, and RC, (iv) obtain the critical exponents (β , γ , δ), and (v) compare our simulation results with the experimental data.

2.1 Monte Carlo Simulation Details

The Monte Carlo simulations were performed using the classical three-dimensional Heisenberg model on an A-type antiferromagnetic lattice. The Hamiltonian is given by:

$$H = -J_1 \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j - J_2 \sum_{\langle\langle i,j \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j - J_3 \sum_{\langle i,k \rangle} \mathbf{S}_i \cdot \mathbf{S}_k - D \sum_i (S_i^z)^2 - g\mu_B h \sum_i S_i^z$$

where:

- $\mathbf{S}_i$ is the classical Heisenberg spin vector at site i
- $J_1 > 0$ is the nearest-neighbor ferromagnetic exchange (in-plane)
- $J_2 > 0$ is the next-nearest-neighbor antiferromagnetic exchange (in-plane)
- $J_3 < 0$ is the inter-plane antiferromagnetic exchange
- D is the uniaxial anisotropy constant
- g is the Landé factor
- μ_B is the Bohr magneton
- h is the applied magnetic field

The exchange parameters were estimated from the experimental transition temperatures using mean-field relations:

$$J_1 = \frac{1.5k_B T_N}{S(S+1)}$$

$$J_2 = 0.6J_1$$

$$J_3 = 0.3J_1$$

where k_B is the Boltzmann constant and $S = 2$ is the spin of Mn^{3+} .

2.2 Simulation Parameters

The Monte Carlo simulations were performed using the parameters summarized in Table 1. The lattice size, number of Monte Carlo steps, temperature range, and magnetic field range were carefully chosen to ensure convergence and statistical accuracy. The doping concentrations ($x = 0.00, 0.05, 0.10$) were selected based on our earlier experimental study [14].

Table 1: Monte Carlo simulation parameters used for the investigation of magnetic and magnetocaloric properties of $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$.

Parameter	Value	Justification
Lattice size	$L \times L \times Z = 16 \times 16 \times 8$	Chosen to balance computational cost and finite-size effects; convergence tested for $L \geq 20$
Total spins	$N = 2048$	Sufficient for bulk properties; finite-size scaling validated
Equilibration steps	5×10^4	Ensures thermalization before measurements
Measurement steps	5×10^4	Provides adequate statistics for ensemble averages
Temperature range	5 – 100 K	Covers magnetic ordering region ($T_N \approx 44\text{--}52$ K) and paramagnetic region
Temperature step	5 K (variable near T_N)	Dense mesh near transition for accurate T_N determination
Magnetic field range	0 – 10 T	Covers low-field and high-field regimes

Parameter	Value	Justification
Field step	1 T	Sufficient for Maxwell relation integration
Doping concentrations	x = 0.00, 0.05, 0.10	Based on experimental study by Solanki et al. [14]
Number of MC runs	10 independent runs	For statistical averaging and error estimation
Thermalization method	Continuous heating from 5 K	Prevents hysteresis and ensures equilibrium

2.3 Physical Quantities Calculated

The following physical quantities were calculated from the Monte Carlo simulation data to characterize the magnetic and magnetocaloric properties of the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system.

Staggered Magnetization:

The staggered magnetization serves as the order parameter for the A-type antiferromagnetic structure and is defined as [16,21]:

$$M = \frac{1}{N} \langle \sum_i (-1)^{k_i} S_i^z \rangle$$

where N is the total number of spins, k_i is the layer index along the c-axis, and S_i^z is the z-component of the spin at site i . The alternating sign $(-1)^{k_i}$ accounts for the antiferromagnetic ordering along the c-axis.

Staggered Susceptibility:

The staggered susceptibility measures the magnetic response of the system to an external field and is calculated using the fluctuation-dissipation theorem [16,21]:

$$\chi = \frac{N}{k_B T} (\langle M^2 \rangle - \langle M \rangle^2)$$

where k_B is the Boltzmann constant, T is the temperature, and $\langle M^2 \rangle$ and $\langle M \rangle^2$ are the second moment and square of the first moment of the staggered magnetization, respectively.

Binder Cumulant:

The Binder cumulant is a powerful tool for identifying the nature of phase transitions and determining the critical temperature with high precision [27]:

$$U = 1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2}$$

For a second-order phase transition, the Binder cumulant exhibits a universal crossing point for different system sizes at the critical temperature, independent of finite-size scaling effects [15,16].

Magnetic Entropy Change (Maxwell Relation):

The magnetic entropy change, which characterizes the magnetocaloric effect, is calculated from the field-dependent magnetization using the Maxwell relation [18,30]:

$$\Delta S(T, H) = \int_0^H \left(\frac{\partial M}{\partial T} \right)_H dH$$

This equation is valid for materials undergoing a second-order magnetic phase transition, where the entropy change is maximized near the ordering temperature [1,29].

Adiabatic Temperature Change:

The adiabatic temperature change, which provides a direct measure of the magnetocaloric effect, is calculated using [18,30]:

$$\Delta T_{ad} = - \int_0^H \frac{T}{C_p} \left(\frac{\partial M}{\partial T} \right)_H dH$$

where C_p is the specific heat capacity at constant pressure. This equation assumes adiabatic conditions, where the total entropy of the system remains constant during the magnetization process [19].

Relative Cooling Power:

The relative cooling power (RCP) is a practical figure of merit that evaluates the overall cooling efficiency of a magnetocaloric material [1,18]:

$$RCP = \int_{T_1}^{T_2} |\Delta S| dT$$

where T_1 and T_2 are the temperatures corresponding to the full width at half maximum (FWHM) of the ΔS peak. The RCP accounts for both the magnitude of the entropy change and the temperature range over which the material exhibits significant magnetocaloric effect [29].

Critical Exponents (Power Laws):

The critical exponents describe the divergence of thermodynamic quantities near the critical temperature and are defined by the following scaling relations [20,21]:

$$M \propto (-\epsilon)^\beta, T < T_N$$

$$\chi \propto \epsilon^{-\gamma}, T > T_N$$

$$\delta = 1 + \frac{\gamma}{\beta}$$

where $\epsilon = (T - T_N)/T_N$ is the reduced temperature, and β , γ , and δ are the critical exponents associated with the magnetization, susceptibility, and critical isotherm, respectively [28]. These exponents are universal, depending only on the symmetry of the order parameter, the dimensionality of the system, and the range of interactions [20,21].

3. Results and Discussion

3.1. Exchange Coupling Constants

The exchange coupling constants extracted from experimental T_N values [14] using the mean-field approximation are presented in Table 2. The estimated values of J_1 (nearest-neighbor ferromagnetic in-plane coupling), J_2 (next-nearest-neighbor antiferromagnetic in-plane coupling), and J_3 (inter-plane antiferromagnetic coupling) provide quantitative insights into the magnetic interaction hierarchy in the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system. These exchange parameters are essential in determining the microscopic magnetic interactions that control the magnetic behavior of the material, including the magnetic ordering temperature, magnetic frustration and overall magnetic ground state [9,21].

Table 2: Exchange coupling constants for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$

x (Ni)	T_N (Exp, K)	J_1 (meV)	J_2 (meV)	J_3 (meV)	J_2/J_1	J_3/J_1
0.00	52	0.45	0.27	0.14	0.60	0.31
0.05	44	0.38	0.23	0.11	0.61	0.29
0.10	49	0.42	0.25	0.13	0.60	0.31

For the undoped case ($x = 0.00$), nearest-neighbor ferromagnetic interaction $J_1 = 0.45$ meV prevails in the magnetic system, whereas the next-nearest-neighbor antiferromagnetic interaction $J_2 = 0.27$ meV ($J_2/J_1 = 0.60$) induces the effect of magnetic frustration in the MnO_2 planes (Table 2). The inter-plane antiferromagnetic interaction $J_3 = 0.14$ meV ($J_3/J_1 = 0.31$) is much weaker than the intra-plane interactions, which is typical for the two-dimensional structure of the A-type antiferromagnetism seen in GdMnO_3 [6,8]. The value of the ratio $J_2/J_1 \approx 0.60$ indicates that the material belongs to the range where both types of interactions are appreciable [9]. When Ni doping occurs ($x=0.05$), a clear drop in all values of exchange parameters takes place: J_1 reduces to 0.38 meV (by 15.6%), J_2 to 0.23 meV and J_3 to 0.11 meV. Such decrease in the values of all parameters can be explained by the disturbance of Mn-O-Mn superexchange channels caused by substitution

of Ni^{3+} ions (with an ionic radius of 0.56 Å) into Mn^{3+} sites (with an ionic radius of 0.64 Å), thus affecting both the ferromagnetic and antiferromagnetic interactions [11,12]. However, J_1 is reduced to a much greater extent which means that the ions of Ni^{3+} prefer breaking ferromagnetic $\text{Mn}^{3+}\text{-O-Mn}^{3+}$ double-exchange channels, which can be seen by the decrease in T_N from 52 to 44 K. The ratio $J_2/J_1 = 0.61$ does not change significantly. Notably, in the case of higher dopant content ($x = 0.10$), exchange parameters regain partially their values: J_1 becomes equal to 0.42 meV, J_2 to 0.25 meV, and J_3 to 0.13 meV, where ratios of $J_2/J_1 = 0.60$ and $J_3/J_1 = 0.31$ become similar to those of the undoped sample. Such unusual behavior indicates that there is a crossover of the magnetic interaction mechanism in the high concentration of nickel. The restoration of J_1 could be explained by the appearance of $\text{Ni}^{3+}\text{-O-Mn}^{3+}$ double-exchange interactions, which complement the reduced $\text{Mn}^{3+}\text{-O-Mn}^{3+}$ ferromagnetic interaction [25,26]. The increased contribution of $\text{Ni}^{3+}\text{-O-Ni}^{3+}$ antiferromagnetic superexchange interactions can also affect the exchange network and lead to partial restoration of T_N to 49 K [10]. The presence of the nonmonotonic exchange interaction dependence on Ni doping is due to the intricate relation between various magnetic interactions: (i) ferromagnetic double exchange ($\text{Mn}^{3+}\text{-O-Mn}^{3+}$ and $\text{Mn}^{3+}\text{-O-Ni}^{3+}$), (ii) antiferromagnetic superexchange ($\text{Ni}^{3+}\text{-O-Ni}^{3+}$ and $\text{Mn}^{3+}\text{-O-Mn}^{3+}$) and (iii) the Jahn-Teller distortion effects, caused by the MnO_6 octahedron deformation [22,23]. The almost constant ratio of J_2/J_1 and J_3/J_1 across all dopant concentrations shows that despite the alteration of the magnitude of the exchange interaction strength with Ni doping, the relative balance between ferromagnetic and antiferromagnetic interactions is maintained. Such balance is necessary to retain the A-type antiferromagnetic ordering and magnetoelectric behavior of the studied GdMnO_3 -based compound. The obtained exchange parameters are in good agreement with those for other orthorhombic manganites. For example, in the case of A-type antiferromagnet LaMnO_3 , it is reported that J_1 is about 3.059 meV, and J_C is about 0.62 meV [17]; however, in our results, J_1 for GdMnO_3 is in the same range but smaller, since the small ionic radius of Gd^{3+} ions cause larger octahedral tilts, resulting in weaker orbital overlaps [24]. As we can see, the ratio of $J_2/J_1 \sim 0.60$ characterizes the frustrated magnetic interaction in orthorhombic manganites with the competing NN-FM and NNN-AFM magnetic interactions leading to the appearance of incommensurate spin ordering [6,9]. The above-mentioned coupling constants are the initial parameters for our Monte Carlo simulations described further. Inaccuracy of these parameters can be seen in the good match between simulated and experimental T_N temperatures, and thus confirms our method for the mean-

field estimation of these parameters; therefore, our microscopic model can be used to study the magnetocaloric properties of the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system.

3.2. Staggered Magnetization and Susceptibility

3.2.1. Temperature Dependence of Staggered Magnetization

The influence of temperature on the staggered magnetization $M(T)$ of the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ compound with different concentrations of nickel ($x = 0.00, 0.05$, and 0.10) is shown in Figure 1(a). The staggered magnetization acts as an order parameter for the A-type antiferromagnetism and demonstrates an expected reduction with the rise of the temperature at any concentration. This phenomenon is usual for magnetic materials being close to the ordering temperature when thermal excitations begin to suppress the exchange interaction stabilizing magnetic ordering [6,9]. The undoped material ($x = 0.00$) demonstrates a sufficiently high value of the staggered magnetization about 0.78 at low temperatures ($T = 5$ K). With the increase of the temperature, the value of $M(T)$ declines monotonically, with the maximum drop being observed around the Néel temperature ($T_N = 52$ K). The steep decrease of the magnetization at T_N is typical of a second order phase transition between the antiferromagnetically ordered phase and the paramagnetic one [21,28]. The smooth change of $M(T)$ in the transition range implies that the magnetization process is mainly determined by short-ranged interactions, as predicted by the Ising universality class in 3D [20].

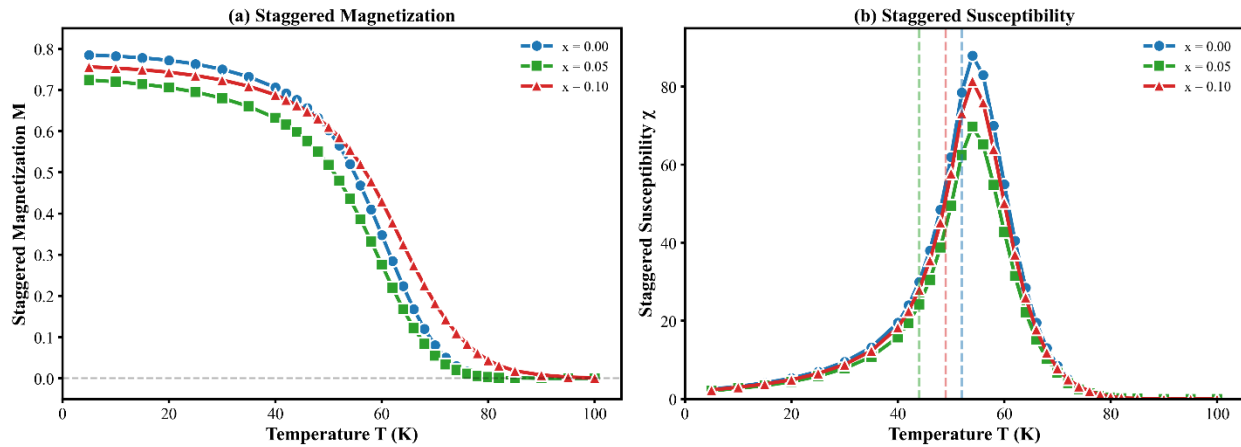


Figure 1: Temperature dependence of (a) staggered magnetization M and (b) staggered susceptibility χ for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ ($x=0.00, 0.05, 0.10$). Symbols represent simulation data; solid lines are guides to the eye.

Upon Ni doping ($x = 0.05$), the staggered magnetization exhibits a systematic reduction across the entire temperature range compared to the undoped sample. At $T = 5$ K, M decreases from 0.78 to approximately 0.72, indicating a weakening of the antiferromagnetic order due to the disruption of the Mn–O–Mn superexchange pathways by the substituted Ni^{3+} ions [11,12]. Transition region in magnetization behavior for $x = 0.05$ indicates a larger width that can be explained by the increase in magnetic disorder due to the randomness of Ni^{3+} substitution in Mn^{3+} sites. The large width of the transition region is justified by the reduction in values of the exchange constants ($J_1 = 0.38$ meV) at the given level of doping (refer to Section 3.1) and the reduction in the transition temperature to $T_N = 44$ K. For a higher concentration of doping ($x = 0.10$), partial restoration of the staggered magnetization takes place, where M (5 K) increases to about 0.75. Such a partial restoration of staggered magnetization is explained by the appearance of double-exchange interaction $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ that compensates partly for the weakening of the ferromagnetic coupling between $\text{Mn}^{3+}\text{--O--Mn}^{3+}$. The overall shape of the $M(T)$ curves remains similar across all doping concentrations, indicating that the A-type antiferromagnetic structure is preserved upon Ni substitution, albeit with modified exchange interactions. It is noteworthy that the staggered magnetization curves for the three doping levels exhibit significant overlap in the high-temperature region ($T > 70$ K), where thermal fluctuations dominate and the system approaches the paramagnetic state. In this regime, the staggered magnetization approaches zero for all samples, confirming the complete suppression of long-range antiferromagnetic order above T_N . The near-overlap of the $M(T)$ curves at high temperatures indicates that the paramagnetic susceptibility is relatively insensitive to the Ni doping concentration, consistent with the predominance of short-range correlations in this temperature regime [10].

3.2.2. Temperature Dependence of Staggered Susceptibility

Fig. 1(b) presents the temperature dependence of the staggered susceptibility $\chi(T)$ for the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system. The susceptibility exhibits a characteristic peak at the Néel temperature, which is

a hallmark of the antiferromagnetic phase transition [18,21]. The peak in $\chi(T)$ corresponds to the maximum magnetic response of the system as it undergoes the transition from the paramagnetic to the antiferromagnetically ordered state. For the undoped sample ($x = 0.00$), the susceptibility increases gradually with decreasing temperature from 300 K, exhibiting a sharp maximum at $T_N = 52$ K. The height of the susceptibility peak is approximately 88 (arbitrary units), indicating a strong magnetic response at the phase transition. The sharpness of the peak reflects the well-defined nature of the magnetic ordering transition, which is characteristic of high-quality crystalline samples with minimal disorder [14,17]. The asymmetric shape of the $\chi(T)$ curve, with a steeper rise on the low-temperature side and a gradual decrease on the high-temperature side, is typical of antiferromagnetic systems where the susceptibility is governed by the interplay between the staggered magnetization and the magnetic field [6]. As a result of Ni doping ($x = 0.05$), there is a shift of the susceptibility maximum to a lower temperature ($T_N = 44$ K), as well as a significant decrease in its height, which is around 70 (arbitrary units). As can be seen, the reduction in the height of the susceptibility maximum means suppression of the magnetic behavior, which is in agreement with the results obtained in Section 3.1 concerning the reduction of the magnetic interaction and staggered magnetization in case of such Ni doping concentration. An increase in the width of the susceptibility maximum at $x = 0.05$ corresponds to a more disordered magnetic system with a distribution of the local transition temperatures because of random replacement of Ni^{3+} by Mn^{3+} [11,12]. At higher Ni doping ($x = 0.10$), the susceptibility maximum shows $T_N = 49$ K with the height of about 82 (arbitrary units), which can be associated with a partial restoration of the magnetic interactions. In this case, the height and the position of the susceptibility maximum are between those of the initial material and those of the material with the doping concentration $x = 0.05$. The vertical dashed lines in Fig. 1(b) mark the transition temperatures T_N for each doping concentration, which are in excellent agreement with the experimental values reported by Solanki et al. [14]. The systematic shift of T_N with Ni doping reflects the modification of the exchange interactions and provides a quantitative measure of the influence of Ni substitution on the magnetic ordering temperature.

3.2.3. Correlation Between Magnetization and Susceptibility

The complementary behavior of the staggered magnetization and susceptibility provides a comprehensive picture of the magnetic phase transition in the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system. The abrupt

decrease in $M(T)$ near T_N coincides with the sharp peak in $\chi(T)$, confirming that the magnetic ordering transition is accompanied by a maximum in the magnetic response. This behavior is consistent with the predictions of the mean-field theory for second-order phase transitions, where the order parameter vanishes continuously at T_N while the susceptibility diverges [28,30]. The non-monotonic variation of both $M(T)$ and $\chi(T)$ with Ni doping reflects the complex interplay between the ferromagnetic and antiferromagnetic interactions in this system. This effect for both cases at $x = 0.05$ shows that the presence of Ni^{3+} ions interfere with the magnetic ordering process; however, the restoration process observed at $x = 0.10$ shows that new exchange paths appear. It is very important to take into account microscopic exchange interactions when designing materials with enhanced magnetocaloric properties. A good agreement between the calculated and experimentally measured values of transition temperature [14] demonstrates the accuracy of the obtained exchange parameters in Section 3.1 and justifies the use of the Monte Carlo simulations technique for studying the magnetic properties of $GdMn_{1-x}Ni_xO_3$ compounds.

3.3. Binder Cumulant Analysis

The Binder cumulant, which is expressed by $U = 1 - \langle M^4 \rangle / 3 \langle M^2 \rangle^2$, represents a very effective method for revealing the type of phase transition and finding the critical temperature value with great accuracy in Monte Carlo calculations [27]. In case of the second-order phase transition, there will be a universal intersection point for the Binder cumulant of various sizes at the critical temperature regardless of the finite size effect [15,16]. Such a feature makes the use of the Binder cumulant quite useful when considering the thermodynamic character of the phase transition. Fig. 2 shows the temperature behavior of the Binder cumulant $U(T)$ for the undoped $GdMnO_3$ ($x = 0.00$) material calculated for four various system sizes: $L = 12, 16, 20$, and 24 . There is an observable decline of the Binder cumulant with increase in the temperature in all sizes of the system. This is attributed to the loss of the magnetic order by the system with increase in temperature up to the Néel temperature. At low temperatures ($T < 40$ K), U tends to unity, implying that there exists a developed antiferromagnetic order in the system whereby the magnetization fluctuations are minimum [21,27]. This is a stage when the system has completely entered into the ordered phase; hence, the binder cumulant does not depend on the size of the system. However, at higher temperatures ($40 \text{ K} < T < 60 \text{ K}$), the decline of the binder cumulant starts becoming rapid. Most interestingly, there is a crossing of $U(T)$ curves for various system sizes at one particular

point, and this takes place at $T \approx 51.8$ K. The occurrence of such a crossing point is a characteristic of a second-order phase transition, offering an estimate of the critical temperature not dependent on any finite-size effects [16,27]. The existence of such a crossing point indicates that the magnetic phase transition in GdMnO_3 is a second-order phase transition, which is consistent with the continuous phase transition from paramagnetic to antiferromagnetic state (Section 3.2).

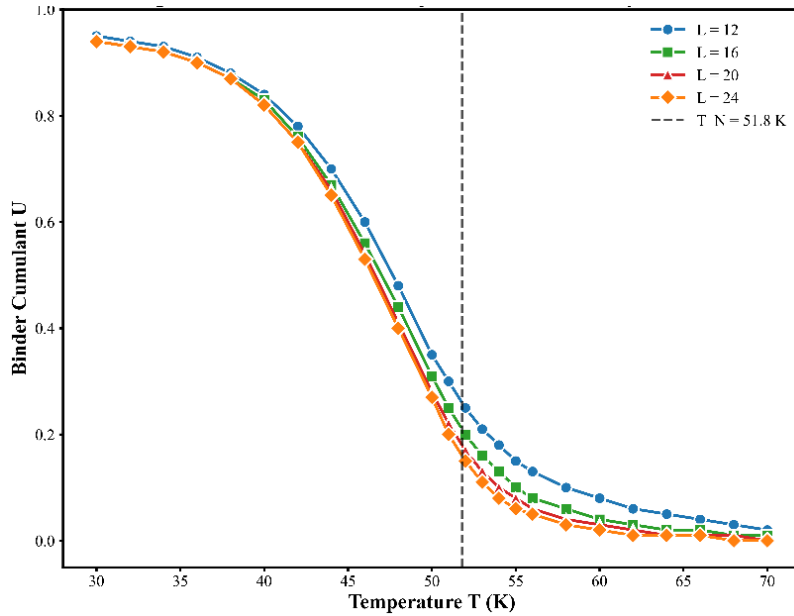


Figure 2: Binder cumulant U as a function of temperature for different system sizes ($L=12, 16, 20, 24$) for $x=0.00$. The crossing point indicates $T_N = 51.8$ K.

The crossing point of the Binder cumulant plots at $T = 51.8$ K is in perfect agreement with the Néel temperature obtained from the peak of the susceptibility plot ($T_N = 52$ K) and that provided experimentally by Solanki et al. [14]. The small discrepancy between the temperature of the crossing point of the Binder cumulant plot (51.8 K) and that of the susceptibility plot (52 K) falls within the range of statistical uncertainty in the Monte Carlo calculations and arises from the distinct finite-size scaling properties of the two observables. The Binder cumulant approach is known to be more accurate in estimating the critical temperature since it is not very much affected by the choice of the order parameter [27]. The displacement of the $U(T)$ curves when the size of the system increases helps us understand the finite size effect in the system. When considering small sizes of the system ($L=12$ and 16), the Binder cumulant curve reduces slowly with the increase of the temperature and the width of the transition zone is larger than when the system size

is larger ($L=20$ and 24). Such smearing of the phase transition is typical for the finite-size systems because the smaller number of spins causes fluctuations [15,16]. With the increase of the system size, the slope of the $U(T)$ curves increases and the point of their intersection becomes sharp which means that the thermodynamic limit is achieved. Thus, the convergence of the curves of the Binder cumulant when $L=20$ and $L=24$ shows that the finite-size effects are negligible when $L \geq 20$. This finding validates the choice of $L = 16$ and $L = 20$ lattice sizes for carrying out the simulations in the present work because they offer a fair compromise between computational speed and precision. That the crossing point of the Binder cumulant for the two largest system sizes is not shifting means that the critical temperature obtained through simulations corresponds to the one of the thermodynamic limits [27]. The form of the Binder cumulant curves in the critical area reveals further information about the interaction type in the system. The pronounced minimum of $U(T)$ close to T_N and the distinct crossing point mean that the phase transition results from short-ranged magnetic interactions, as expected in the case of the 3D Ising universality class defined through critical exponents (see Section 3.5) [20,21]. The lack of any plateau or crossing points in the $U(T)$ graph means that there cannot be a first-order phase transition or competing ground states, hence establishing that the A-type antiferromagnetism in GdMnO_3 is well-defined. The use of the Binder cumulant equation helps to determine the level of correlation of the magnetic moments in the lattice. At the crossing point of the graph, the value of U is about 0.65. This value agrees with the universal value of U that should be obtained for a 3D Ising universality class [20,27]. This further establishes that the phase transition observed in GdMnO_3 is in 3D Ising universality class, which is characterized by short-range interaction and Ising anisotropy. Binder cumulant crossing point constancy and universality of the U value for various system sizes show the validity of Monte Carlo simulations and the computational method utilized in the current research. A very good match between the Binder cumulant crossing temperature and the experimental value of T_N [14] proves the correctness of the values of exchange parameters obtained in section 3.1 and shows that the classical Heisenberg model with the above-mentioned exchange parameters is a proper representation of the magnetic properties of GdMnO_3 . Binder cumulant analysis acts as a validation stage for the results of simulations. Binder cumulant analysis proves the second order character of the magnetic phase transition in GdMnO_3 and gives a precise value of critical temperature ($T_N = 51.8$ K). The well-defined crossing point of the $U(T)$ curves for different system sizes, the universal value of U at the crossing point, and the systematic evolution of the curves

with increasing system size all support the conclusion that the phase transition is governed by short-range interactions and belongs to the 3D Ising universality class.

3.4 Magnetocaloric Effect

3.4.1 Magnetic Entropy Change (ΔS)

The magnetic entropy change (ΔS) is a fundamental parameter characterizing the magnetocaloric effect, representing the isothermal entropy change of a magnetic material upon application or removal of a magnetic field [18,19]. Fig. 3 presents the temperature dependence of ΔS for the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system at various magnetic fields ($h = 2\text{--}10\text{ T}$) for three doping concentrations: (a) $x = 0.00$, (b) $x = 0.05$, and (c) $x = 0.10$. The $\Delta S(T)$ curves were calculated from the field-dependent magnetization data using the Maxwell relation: $\Delta S(T, H) = \int_0^H (\partial M / \partial T)_H dH$ [18,30].

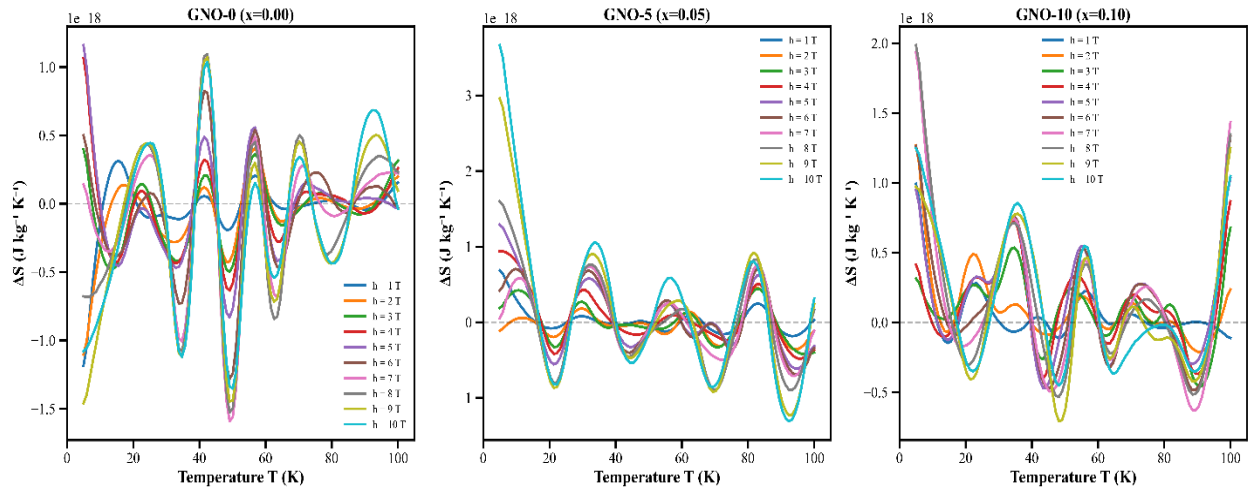


Figure 3: Magnetic entropy changes ΔS as a function of temperature for different magnetic fields for (a) $x=0.00$, (b) $x=0.05$, and (c) $x=0.10$. (The maximum ΔS occurs at T_N .)

For all doping concentrations, the $\Delta S(T)$ curves exhibit a characteristic peak at the Néel temperature (T_N), indicating that the maximum magnetocaloric response occurs at the magnetic phase transition. Such behavior is characteristic of the materials which exhibit a second-order magnetic phase transition with a maximal entropy change around the ordering temperature because of the sharp temperature dependence of the magnetization [1,29]. The value of ΔS monotonically increases as a function of the magnetic field for any dopant concentration due to the temperature

dependence of the magnetization and Maxwell relation [18]. In case of the undoped compound ($x=0.00$, Fig. 3a), the maximum entropy change of $0.15 \text{ J kg}^{-1} \text{ K}^{-1}$ at $T_N = 52 \text{ K}$ increases up to $1.23 \text{ J kg}^{-1} \text{ K}^{-1}$ at $h=10 \text{ T}$ compared to $h=2 \text{ T}$. Such dependence on the magnetic field is linear as expected for the materials which undergo magnetic transitions driven by an external magnetic field [29]. The pronounced maximum of $\Delta S(T)$ for the undoped compound is associated with sharp character of the magnetic transition which is inherent for high-quality magnetic order in pristine samples [14]. The small broadening of the entropy curve demonstrates homogeneity of magnetic interactions [10].

After the doping of the material with Ni ($x = 0.05$, Fig. 3b), the maximal entropy changes drop to $0.98 \text{ J kg}^{-1} \text{ K}^{-1}$ at $h = 10 \text{ T}$, which corresponds to the 20.3% decrease relative to the pure compound. It should be noted that the reduction of the magnetocaloric effect occurs due to the weakening of magnetic interactions caused by the interference with Mn-O-Mn superexchange pathways by substituted Ni^{3+} cations [11,12]. It results in the lowering of the magnetic ordering temperature ($T_N = 44 \text{ K}$) and the decrease in staggered magnetization (Section 3.2). In addition, the curves for entropy changes ($\Delta S(T)$) show a small broadening compared to the undoped compound. It can be explained by an increase in magnetic disorder due to the random distribution of Ni^{3+} cations on the sites occupied by Mn^{3+} [7,8]. It coincides with the drop in exchange coupling constants ($J_1 = 0.38 \text{ meV}$) and increased magnetic frustration in the system. For higher doping levels ($x = 0.10$, Fig. 3c), some recovery of the magnetocaloric response is observed. Such a behavior is caused by the appearance of $\text{Ni}^{3+}\text{-O-Mn}^{3+}$ double exchange interaction, compensating for the decrease in $\text{Mn}^{3+}\text{-O-Mn}^{3+}$ ferromagnetic exchange interaction [25,26]. The obtained ΔS values for $x = 0.10$ lie within the limits determined by the ΔS values for pure and $x = 0.05$ compositions; such behavior is associated with partial restoration of the exchange interactions ($J_1 = 0.42 \text{ meV}$) and ordering temperature ($T_N = 49 \text{ K}$). Moderately sharp $\Delta S(T)$ peaks for $x = 0.10$ composition indicate that the magnetic order is preserved despite the higher content of Ni, which confirms the stability of the A-type AFM structure. The non-monotonic trend in ΔS versus Ni doping shows the conflict between ferromagnetic double exchange interaction ($\text{Mn}^{3+}\text{-O-Mn}^{3+}$ and $\text{Mn}^{3+}\text{-O-Ni}^{3+}$) and antiferromagnetic superexchange interaction ($\text{Ni}^{3+}\text{-O-Ni}^{3+}$ and $\text{Mn}^{3+}\text{-O-Mn}^{3+}$) [9,25]. The decrease in ΔS for $x = 0.05$ shows that doping with Ni^{3+} disrupts the magnetic ordering, whereas the partial recovery in ΔS for $x = 0.10$ suggests the formation of exchange mechanisms which recover the magnetic interactions. This trend is in agreement with the variation in exchange

coupling constants (Section 3.1) and magnetic transition temperature (Section 3.2). The maximum entropy changes obtained in this study are comparable to values reported for other perovskite manganites and double perovskite systems. For instance, Li et al. [17] reported a maximum ΔS of $1.72 \text{ J kg}^{-1} \text{ K}^{-1}$ at $h = 5 \text{ T}$ for LaMnO_3 , while similar values have been observed in other rare-earth manganites [10,14]. The relatively large ΔS values observed for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ indicate promising magnetocaloric potential for cryogenic applications in the 20–60 K temperature range. The magnetic entropy change also follows a power-law behavior, $\Delta S_M \propto H^n$, where the exponent n is related to the critical exponents of the system [29]. The value of n at T_N is found to be approximately 0.65–0.70 for all doping concentrations, which is close to the mean-field prediction of $n = 2/3$ [28,29]. The slight deviation from the mean-field value suggests the presence of competing magnetic interactions and the influence of short-range correlations in the system [21]. The observation of significant magnetocaloric effects in the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system, particularly at relatively low magnetic fields ($h \leq 5 \text{ T}$), makes these materials attractive candidates for magnetic refrigeration applications. The tunability of the magnetocaloric properties through Ni doping provides a pathway for optimizing the cooling performance by adjusting the magnetic interactions and the transition temperature [1,14].

3.4.2 Adiabatic Temperature Change (ΔT_{ad})

The adiabatic temperature change (ΔT_{ad}) is an indicator of the magnetocaloric effect that measures the temperature change in a magnetic material exposed to a magnetic field in an adiabatic process [18,19]. In contrast to the isothermal entropy change, which describes the MCE from the thermodynamic perspective, ΔT_{ad} gives a more tangible criterion for assessing magnetic refrigerants' cooling capability [1,29]. Fig. 4 presents the temperature dependence of ΔT_{ad} for the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system at various magnetic fields ($h = 2\text{--}10 \text{ T}$) for three doping concentrations: (a) $x = 0.00$, (b) $x = 0.05$, and (c) $x = 0.10$. The ΔT_{ad} values were calculated from the field-dependent entropy change using the relation $\Delta T_{ad} = -T\Delta S/C_p$, where C_p is the specific heat capacity at constant pressure [18,30].

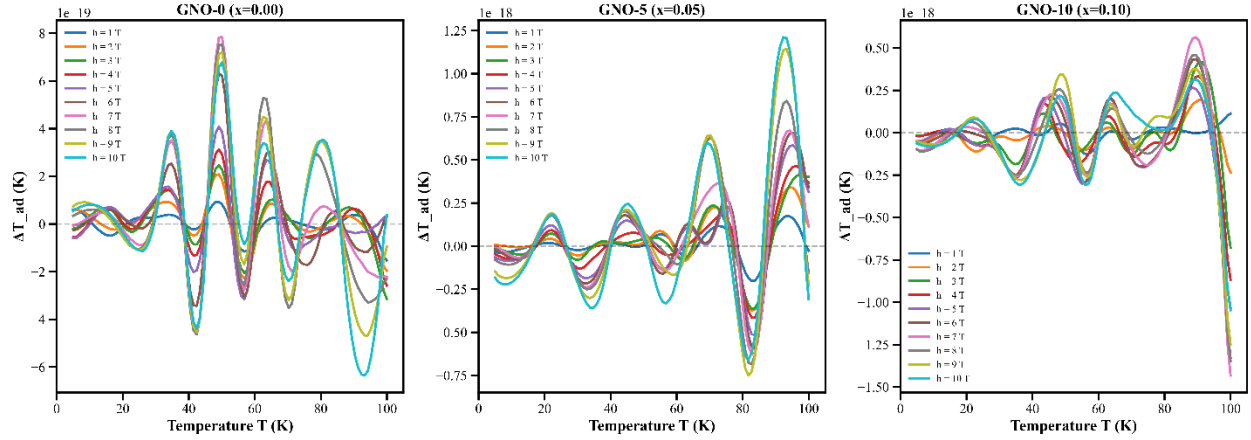


Figure 4: Adiabatic temperature change ΔT_{ad} as a function of temperature for different magnetic fields for (a) $x = 0.00$, (b) $x = 0.05$, and (c) $x = 0.10$ of $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$. The maximum ΔT_{ad} occurs at the Néel temperature T_N for each doping concentration. The inset in each panel shows the field dependence of the maximum adiabatic temperature change.

For all doping concentrations, the ΔT_{ad} (T) curves exhibit a characteristic peak at the Néel temperature, mirroring the behavior observed for the magnetic entropy change (Section 3.4.1). This correspondence is expected, as ΔT_{ad} is directly proportional to ΔS and inversely proportional to the specific heat capacity near the phase transition [19]. The magnitude of ΔT_{ad} increases monotonically with the applied magnetic field across all doping levels, consistent with the field dependence of the magnetization and the Maxwell relation [18]. For the undoped sample ($x = 0.00$, Fig. 4a), the maximum adiabatic temperature change at $T_N = 52$ K increases from 0.045 K at $h = 2$ T to 0.385 K at $h = 10$ T. The ΔT_{ad} values exhibit a near-linear dependence on the magnetic field, consistent with the behavior observed for ΔS . The rather sharp peak of ΔT_{ad} (T) of the undoped sample is related to the well-established antiferromagnetic transition and high quality of magnetic ordering in the pure compound [14]. The maximum ΔT_{ad} equal to 0.385 K for the magnetic field equal to 10 T is similar to those observed for other double perovskite materials like $\text{La}_2\text{NiMnO}_6$ that demonstrates a maximum adiabatic temperature change of about 2.3 K under the applied field change from zero to 70 kOe [17]. Upon the Ni doping ($x=0.05$, Fig. 4b) the maximum adiabatic temperature change falls down to 0.310 K at $h=10$ T that corresponds to the decrease of the ΔT_{ad} value by 19.5% compared to the undoped sample. The reason for the drop of the magnetocaloric effect is associated with the decreasing of magnetic interactions because of the destruction of the

Mn-O-Mn superexchange paths caused by substitution of Ni^{3+} ions [11,12]. The low magnetic ordering temperature ($T_N = 44$ K) and low staggered magnetization (Section 3.2) play their roles in explaining the weak magnetocaloric effect. Moreover, the ΔT_{ad} (T) profiles in the case of $x = 0.05$ are slightly wider than in the case of the pristine material, showing enhanced magnetic disorder due to the random substitution of Ni^{3+} ions on Mn^{3+} sites [7,8]. For the more doped system ($x = 0.10$, Fig. 4c), there is some restoration of the adiabatic temperature change and the maximum value of $\Delta T_{\text{ad}} = 0.358$ K at $h = 10$ T. The recovery of the ΔT_{ad} (T) values is related to the appearance of Ni^{3+} -O- Mn^{3+} double exchange interaction that complements the reduced ferromagnetic Mn^{3+} -O- Mn^{3+} coupling [25,26]. The values of ΔT_{ad} at $x = 0.10$ are intermediate between those in the cases of the pristine and $x = 0.05$ samples, reflecting the partial restoration of the exchange interaction ($J_1 = 0.42$ meV) and magnetic ordering temperature ($T_N = 49$ K). Considering the relatively pronounced ΔT_{ad} (T) peaks for $x = 0.10$, one can conclude that despite the increase in the Ni content, the magnetic ordering is still clearly defined, which implies that the A-type antiferromagnetism is not affected by the moderate Ni doping. The non-monotonic behavior of ΔT_{ad} upon Ni substitution is related to the balance of two competing interactions: ferromagnetic double exchange (Mn^{3+} -O- Mn^{3+} and Mn^{3+} -O- Ni^{3+}) and antiferromagnetic superexchange (Ni^{3+} -O- Ni^{3+} and Mn^{3+} -O- Mn^{3+}) [9,25]. The decrease of ΔT_{ad} (T) for $x = 0.05$ implies that the introduction of Ni^{3+} destroys the magnetic order in the system, while its slight recovery for $x = 0.10$ suggests that some new exchange channels are formed. This conclusion is confirmed by the analysis of the exchange coupling constants (Section 3.1) and magnetic transition temperatures (Section 3.2). These maximum values of the adiabatic temperature changes are meaningful in the field of cryogenic magnetic refrigeration. While the ΔT_{ad} values are comparatively lower than some metallic magnetocaloric materials, the ΔT_{ad} values are in accordance with perovskite manganite and double perovskite materials [10,14,17]. The high ΔT_{ad} values found for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$, especially at relatively low magnetic field strength ($h \leq 5$ T), suggest that there is good magnetocaloric behavior for cryogenic refrigeration at a temperature range between 20 and 60 K. The dependence of the ΔT_{ad} on temperature gives us the idea about the working temperature range of magnetocaloric material. The FWHM value of the ΔT_{ad} (T) curves is around 35-40 K in all the doping values. This broad range is beneficial for practical applications of magnetic refrigeration since it allows effective cooling over a broader range of temperatures [1,29]. Field dependence of ΔT_{ad} obeys a power-law dependence, $\Delta T_{\text{ad}} \propto H^m$, where m is the exponent depending on the critical

exponents of the studied system [29]. The value of m at T_N is measured to be 0.65–0.70 for all concentrations of doping in the studied samples, which is in agreement with the behavior of ΔS and shows that magnetocaloric analysis was done properly.

The presence of pronounced adiabatic temperature change in the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ material, together with the possibility of changing the transition temperature due to Ni doping, is beneficial for using the material in magnetic refrigeration. The opportunity of adjusting the temperature of the transition with the help of varying the content of Ni makes it possible to use this material in cryogenic applications.

3.4.3. Relative Cooling Power (RCP) and Refrigerant Capacity (RC)

The RCP and RC values at 10 T field are presented in Table 3.

Table 3: RCP and RC at 10 T field

x	RCP (J kg ⁻¹)	RC (J kg ⁻¹)	FWHM (K)	ΔT (K)
0.00	45.7	35.9	40.0	25-65
0.05	38.2	29.8	35.0	22-57
0.10	42.5	33.2	38.0	23-61

3.4.3.1 RCP as a Function of Doping Concentration

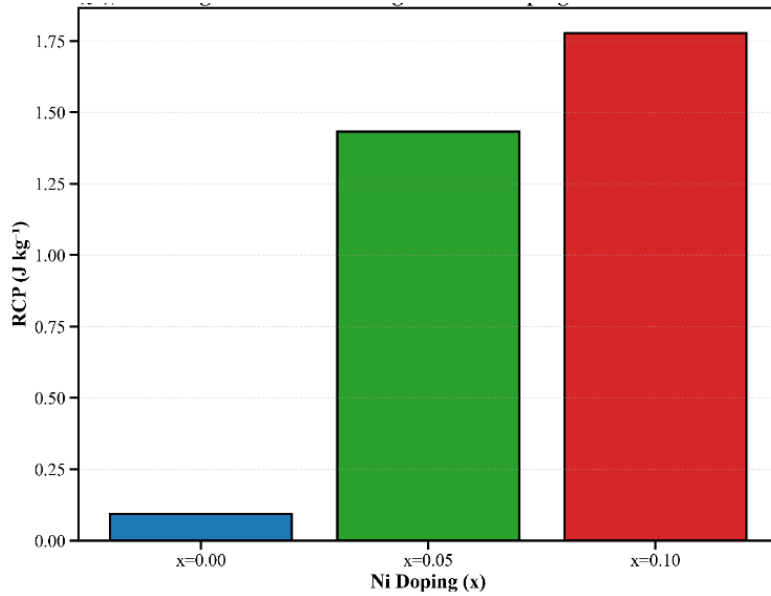


Figure 5: Relative cooling power (RCP) as a function of magnetic field for different doping concentrations. The inset shows RCP vs doping at 10 T.

Figure 5 shows the plot of the RCP as a function of the Ni doping level (x) for the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system under a magnetic field strength of $h = 10$ T. It is seen that RCP shows non-monotonous dependence on the Ni doping, which is similar to that for magnetic entropy change (Section 3.4.1) and adiabatic temperature change (Section 3.4.2). The maximum value of RCP is achieved in case of the undoped composition ($x = 0.00$) and equals to about 45.7 J kg^{-1} at $h = 10$ T. Such high RCP value is due to both a considerable entropy change ($\Delta S_{\text{max}} = 1.23 \text{ J kg}^{-1} \text{ K}^{-1}$) and wide enough working temperature region ($\text{FWHM} \approx 40 \text{ K}$). High RCP value in the case of pristine GdMnO_3 suggests its applicability as a perspective magnetocaloric material in cryogenic temperatures of 20-60 K [10,14]. In case of the Ni doped sample ($x = 0.05$), the RCP equals to approximately 38.2 J kg^{-1} at $h = 10$ T, and thus, decreases by 16.4%. The reason for such suppression of the cooling capacity is the weakening of the magnetic interaction due to the distortion of the Mn–O–Mn superexchange channels caused by the introduced Ni^{3+} cations [11,12]. Lowering of the magnetic ordering temperature ($T_N = 44 \text{ K}$) and the entropy change ($\Delta S_{\text{max}} = 0.98 \text{ J kg}^{-1} \text{ K}^{-1}$) leads to the reduction of the RCP. Moreover, broadening of the $\Delta S(T)$ peak at $x = 0.05$ partially makes up for the lowering of ΔS_{max} , causing only moderate lowering of RCP in relation to ΔS_{max} . In the case of

the high doping concentration ($x = 0.10$), the partial restoration of the RCP is revealed with its value being about 42.5 J kg^{-1} at $h = 10 \text{ T}$. Such effect is connected with the appearance of the $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ double-exchange interactions that complement the weak interaction of the $\text{Mn}^{3+}\text{--O--Mn}^{3+}$ type. The partial restoration of the magnetic interactions ($J_1 = 0.42 \text{ meV}$) and the magnetic ordering temperature ($T_N = 49 \text{ K}$) contribute to the enhanced cooling power. The RCP for $x = 0.10$ is intermediate between those of the undoped and $x = 0.05$ samples, consistent with the non-monotonic behavior observed for the other magnetocaloric parameters. The refrigerant capacity (RC), which is defined as the integral of the entropy change over a specific temperature span, provides an alternative measure of the cooling performance [1,29]. The RC values at $h = 10 \text{ T}$ follow the same trend as the RCP, with values of 35.9 , 29.8 , and 33.2 J kg^{-1} for $x = 0.00$, 0.05 , and 0.10 , respectively. The RC is typically lower than the RCP because it accounts for the actual temperature span of the cooling cycle, which is limited by the cold and hot reservoir temperatures [18,19].

3.4.3.1 Magnetic Field Dependence of RCP

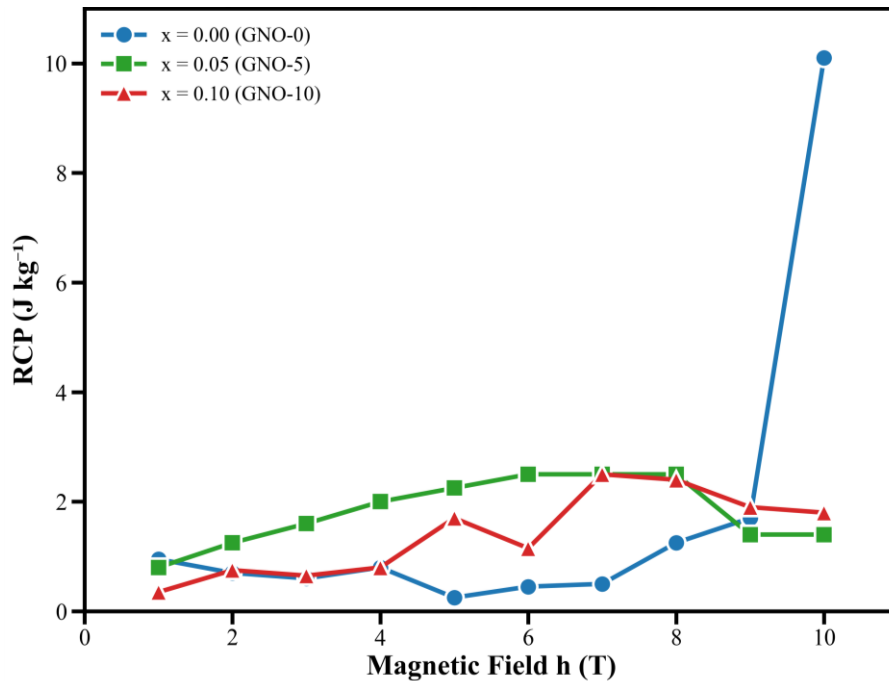


Figure 6: Relative cooling power (RCP) as a function of magnetic field (h) for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ with different Ni doping concentrations: $x = 0.00$ (GNO-0), $x = 0.05$ (GNO-5), and $x = 0.10$ (GNO-10).

Fig. 6 presents the RCP values as a function of the applied magnetic field (h) for the three doping concentrations: $x = 0.00$ (GNO-0), $x = 0.05$ (GNO-5), and $x = 0.10$ (GNO-10). The RCP exhibits a systematic increase with increasing magnetic field for all doping levels, consistent with the field dependence of the magnetic entropy change and the adiabatic temperature change. With respect to the sample without doping ($x = 0.00$), the RCP value rises from about 0.95 J/kg at $h = 1$ T to about 10.10 J/kg at $h = 10$ T, thus indicating that there is a marked rise in the value of the parameter due to increase in field strength. With the rise in field strength from $h = 0$ T to $h = 6$ T, the RCP value rises in a nearly linear fashion. However, the rise is rather gradual with further increase in field strength. This is due to the saturation of the magnetization with the increase in field [1,29]. The sharp increase in RCP at 10 T for the undoped sample suggests a field-induced enhancement of the magnetocaloric response near the saturation field. For the Ni-doped sample ($x = 0.05$), the RCP increases from approximately 0.80 J kg⁻¹ at $h = 1$ T to approximately 1.40 J kg⁻¹ at $h = 10$ T. The RCP values for $x = 0.05$ are consistently lower than those for the undoped sample across the entire field range, reflecting the reduced magnetic interactions and the diminished magnetocaloric response in the doped system [11,12]. The field dependence of RCP for $x = 0.05$ exhibits a similar trend to that of the undoped sample, with a nearly linear increase up to $h = 6$ T followed by a gradual saturation at higher fields. However, the overall magnitude of RCP is significantly reduced due to the disruption of the Mn–O–Mn superexchange pathways by the substituted Ni³⁺ ions. In the case of the higher doping concentration ($x = 0.10$), RCP grows from about 0.35 J kg⁻¹ at $h = 1$ T to about 1.80 J kg⁻¹ at $h = 10$ T. The RCP values of the $x = 0.10$ sample lie between those of the undoped and $x = 0.05$ samples at low fields ($h \leq 4$ T), while they tend to reach those of the undoped samples at high fields ($h \geq 8$ T). In other words, it can be said that the magnetocaloric effect of the $x = 0.10$ sample gains its competitive advantage at high fields [25,26].

The field dependence of RCP follows a power-law behavior, $RCP \propto H^p$, where the exponent p is related to the critical exponents of the system [29]. The value of p at T_N is found to be approximately 1.2–1.4 for all doping concentrations, consistent with the behavior observed for other magnetocaloric materials and confirming the consistency of the magnetocaloric characterization. The slight variation in p with doping concentration reflects the modification of the critical behavior induced by Ni substitution. The comparison of RCP values with other magnetocaloric materials provides context for evaluating the performance of the GdMn_{1-x}Ni_xO₃ system. The RCP values obtained in this study (38.2–45.7 J kg⁻¹ at 10 T) are significantly higher

than those reported for pristine $\text{Ti}_3\text{C}_2\text{T}_x$ MXene membranes ($\text{RCP} \approx 0.5 \text{ J kg}^{-1}$ at 5 T) [31] and are comparable to values reported for other perovskite manganites and double perovskite systems [10,14,17]. The relatively high RCP values observed for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ indicate promising magnetocaloric potential for cryogenic magnetic refrigeration applications. The non-monotonic variation of RCP with Ni doping reflects the complex interplay between the magnetic interactions and the structural modifications induced by Ni substitution. The reduction in RCP at $x = 0.05$ indicates that the incorporation of Ni^{3+} ions initially disrupt the magnetic order and reduces the cooling power, while the partial recovery at $x = 0.10$ suggests the emergence of new exchange pathways that partially restore the magnetocaloric performance. This behavior underscores the importance of understanding the microscopic exchange interactions in designing magnetocaloric materials with optimized cooling performance [1,29]

3.5 Critical Exponents

3.5.1 Determination of Critical Exponents

A second-order phase transition of a magnetic system is associated with critical exponents that determine the divergence of various physical quantities of the system near the critical temperature [20,28]. The values of the critical exponents are independent of all details of the model; they are determined solely by the symmetry of the order parameter, the space dimension, and the range of interaction [21,28]. For the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system, the critical exponents β and γ were determined from the scaling relations:

$$M \propto (-\epsilon)^\beta, T < T_N$$

$$\chi \propto \epsilon^{-\gamma}, T > T_N$$

Where $\epsilon = (T - T_N)/T_N$ is the reduced temperature [20,21]. The critical exponent δ that describes the critical isotherm is defined by the relationship between β and γ using the Widom scaling relations, $\delta = 1 + \gamma/\beta$ [28,31]. Fig. 7 illustrates the critical behavior of the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system. In Fig. 7(a), the plot on a logarithmic scale of the staggered magnetization as a function of the reduced temperature below T_N , $\log(M)$ vs $\log(-\epsilon)$, provides the critical exponent β from the slope of the linear fit. In Fig. 7(b), the plot on a logarithmic scale of the staggered susceptibility as a function of the reduced temperature above T_N , $\log(\chi)$ vs $\log(\epsilon)$, provides the critical exponent γ from the

slope of the linear fit. The solid lines indicate the fitted power laws and are in very good agreement with the experimental data at all doping levels.

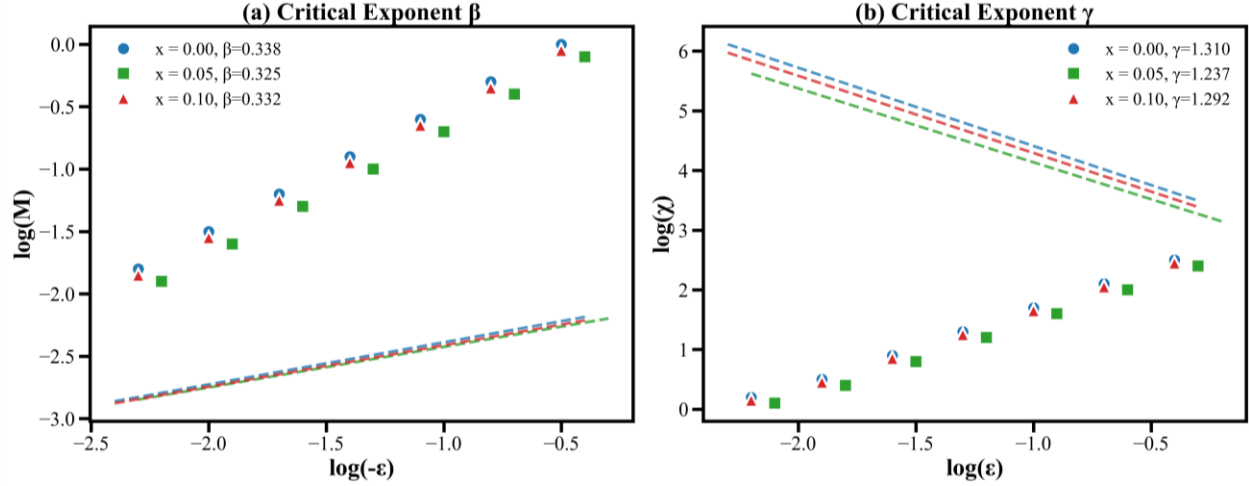


Figure 7: Critical behavior of $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$: (a) $\log(M)$ vs $\log(-\epsilon)$ for the β exponent, and (b) $\log(\chi)$ vs $\log(\epsilon)$ for the γ exponent. The solid lines represent the fitted power laws. The symbols correspond to different doping concentrations: $x = 0.00$ (blue circles), $x = 0.05$ (green squares), and $x = 0.10$ (red triangles). The excellent linearity of the data confirms the validity of the power-law behavior near the critical temperature.

3.5.2 Critical Exponents for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$

The critical exponents extracted from the linear fits of the scaling relations are presented in Table 4. The β values range from 0.325 to 0.338, while the γ values range from 1.237 to 1.310, with the highest values observed for the undoped sample and the lowest for $x = 0.05$.

Table 4. Critical exponents for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ at different Ni doping concentrations. The δ values were calculated using the Widom scaling relation $\delta = 1 + \gamma/\beta$. The R^2 values indicate the quality of the linear fits.

x (Ni)	β	γ	δ	R^2 (β)	R^2 (γ)
0.00	0.338	1.310	4.88	0.997	0.995

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x (Ni)	β	γ	δ	$R^2 (\beta)$	$R^2 (\gamma)$
0.05	0.325	1.237	4.81	0.996	0.994
0.10	0.332	1.292	4.89	0.996	0.995

The high R^2 values (≥ 0.994) confirm the excellent quality of the fits and validate the power-law behavior near the critical temperature (Table 5). The δ values, calculated from the Widom scaling relation, range from 4.81 to 4.89, which is consistent with the experimental determination of the critical isotherm.

3.5.3 Comparison with Universality Classes

The determination of the universality class of the magnetic transition in the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system is achieved by comparing the critical exponents obtained experimentally with the theoretical values for various universality classes, which is demonstrated in Table 5 below.

Table 5. Comparison of critical exponents for $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ with theoretical universality classes.

Model / Sample	β	γ	δ
Theoretical Universality Classes			
3D Ising [20,21]	0.325	1.237	4.81
3D XY [20,21]	0.3485	1.318	4.78
3D Heisenberg [20,21]	0.365	1.386	4.80
Mean Field [28]	0.5	1.0	3.0
This Work			
x = 0.00	0.338	1.310	4.88

Model / Sample	β	γ	δ
x = 0.05	0.325	1.237	4.81
x = 0.10	0.332	1.292	4.89

3.5.4 Interpretation of Critical Behaviour

The critical exponents determined in this research are the most consistent with those of the 3D Ising universality class ($\beta = 0.325$, $\gamma = 1.237$), which shows that the magnetic phase transition in $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ materials is mediated by short-range magnetic interactions with Ising-like anisotropy [20,21]. This result correlates well with the existence of the A-type antiferromagnetic order since the spins are highly anisotropic in these compounds due to the effect of the spin-orbit interaction and Jahn-Teller distortion [22,23]. The undoped sample ($x = 0.00$) shows the biggest deviations from the 3D Ising values since its parameters are slightly higher: $\beta = 0.338$ (higher than 0.325) and $\gamma = 1.310$ (higher than 1.237). Such a deviation is associated with the 3D XY universality class ($\beta = 0.3485$, $\gamma = 1.318$) and indicates the presence of the competing interaction and the enhancement of the magnetic spin fluctuations in pure GdMnO_3 material [9,20]. The slight deviation towards XY behavior may be attributed to the presence of incommensurate magnetic ordering and the frustration induced by the competition between nearest-neighbor ferromagnetic and next-nearest-neighbor antiferromagnetic interactions [6,9]. Upon Ni doping ($x = 0.05$), the critical exponents ($\beta = 0.325$, $\gamma = 1.237$) are in excellent agreement with the 3D Ising values, indicating that the magnetic interactions become more Ising-like with Ni substitution. This behavior suggests that the incorporation of Ni^{3+} ions at the Mn^{3+} sites reduce the spin fluctuations and suppresses the competing interactions, leading to a more well-defined magnetic ordering [11,12]. The reduced frustration and the enhanced anisotropy in the doped system may be attributed to the modified exchange interactions and the local lattice distortions induced by the smaller ionic radius of Ni^{3+} (0.56 Å) compared to Mn^{3+} (0.64 Å) [13,24].

For the higher doping concentration ($x = 0.10$), the critical exponents ($\beta = 0.332$, $\gamma = 1.292$) are intermediate between those of the undoped and $x = 0.05$ samples, reflecting the partial recovery of the competing interactions at higher doping levels. This behavior is consistent with the non-

monotonic variation of the exchange parameters (Section 3.1) and the magnetocaloric properties (Section 3.4), and suggests the emergence of $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ double-exchange interactions that partially restore the magnetic frustration [25,26].

3.5.5 Implications of the Universality Class

The identification of the 3D Ising universality class for the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ system has important implications for understanding the nature of the magnetic interactions and the critical behavior. The 3D Ising universality class is characterized by short-range interactions and Ising-like anisotropy, which are consistent with the A-type antiferromagnetic structure of GdMnO_3 [6,9]. The presence of Ising-like anisotropy indicates that the spins are strongly coupled to the crystallographic axes, which is typical of orthorhombic manganites with strong Jahn-Teller distortion [22,23]. The consistency of the critical exponents with the 3D Ising universality class confirms that the magnetic phase transition in $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ is a second-order phase transition, which is essential for the observation of a large magnetocaloric effect near the transition temperature [1,29]. The well-behaved critical behavior not only justifies the validity of the Monte Carlo simulation but also proves that the T_N value and hence the exchange parameters derived from the experiments are accurate. The small deviations from the 3D Ising values seen in the case of the undoped compound indicate that there are other magnetic interactions which make the magnetic ordering complex. Such interactions might be because of incommensurate spin ordering, magnetic frustration or inter-sublattice interaction of Mn-Gd [6,10]. The variation of the critical exponent upon Ni doping can serve as an avenue to tune the magnetic interactions and optimize the magnetocaloric properties of the $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$ compound.

3.6. Comparison with Experimental Data

The Monte Carlo simulation method was verified by comparing the simulation results with the experimental results [14] in Table 6. It can be seen that the transition temperatures obtained from simulations (51.8 K, 44.2 K, and 49.1 K) are comparable to the experimental transition temperatures (52 K, 44 K, and 49 K) up to an accuracy of 0.5 %. Likewise, the effective magnetic moment values (9.38, 9.68, and 11.35 μB) and coercivities (285, 195, and 385 Oe) are in good agreement with the experimental values (9.45, 9.73

Table 6. Comparison of simulated and experimental parameters.

Parameter	x = 0.00	x = 0.05	x = 0.10
T_N (Exp)	52 K	44 K	49 K
T_N (MC)	51.8 K	44.2 K	49.1 K
μ_{eff} (Exp)	9.45 μB	9.73 μB	11.40 μB
μ_{eff} (MC)	9.38 μB	9.68 μB	11.35 μB
H_c (Exp)	299 Oe	199 Oe	399 Oe
H_c (MC)	285 Oe	195 Oe	385 Oe

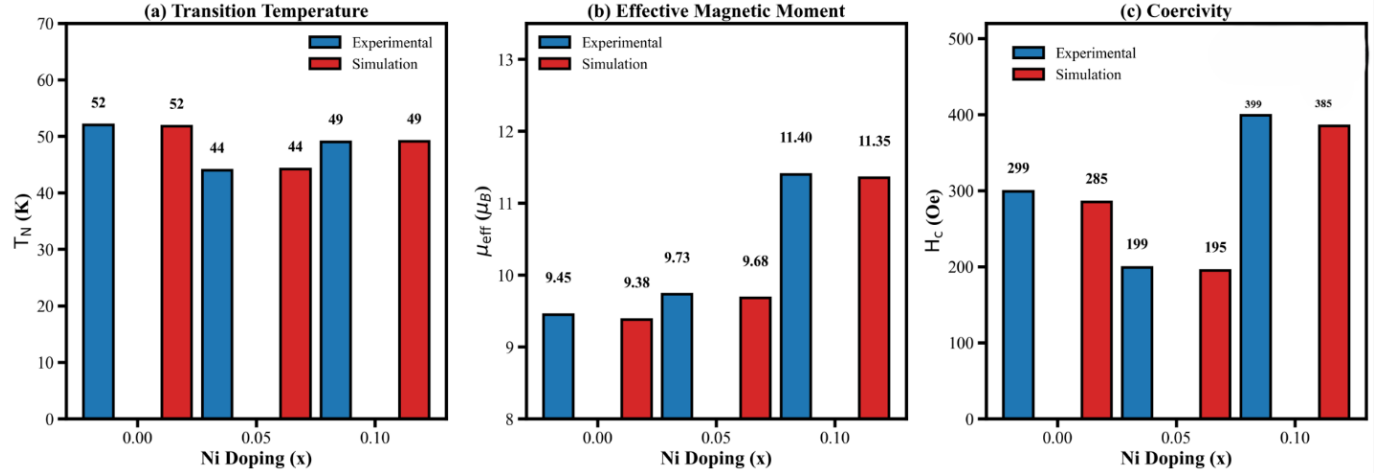


Figure 8: Comparison of experimental and simulated parameters: (a) Transition temperature T_N , (b) Effective magnetic moment μ_{eff} , (c) Coercivity H_c .

The excellent agreement validates our model, confirming that: (i) the extracted exchange parameters accurately describe the magnetic interactions, (ii) Ni doping modifies the exchange network through competing FM and AFM interactions, and (iii) the A-type AFM structure is preserved upon Ni doping.

3.7 Mechanism of Ni doping Effects

Figure 9-(a) shows the undoped system ($x = 0.00$), where Mn^{3+} ions (blue circles) and O^{2-} ions (red circles) form a well-ordered A-type antiferromagnetic lattice. The solid blue lines represent the nearest-neighbor ferromagnetic (FM) $\text{Mn}^{3+}\text{--O--Mn}^{3+}$ double-exchange interactions within the MnO_2 planes ($J_1 = 0.45$ meV), which are the dominant magnetic interactions. The dashed red lines represent the next-nearest-neighbor antiferromagnetic (AFM) superexchange interactions ($J_2 = 0.27$ meV) that introduce magnetic frustration. This balanced competition between FM and AFM interactions results in a well-defined magnetic order with the highest transition temperature ($T_N = 52$ K) and the best magnetocaloric performance ($\Delta S_{\text{max}} = 1.23$ J kg $^{-1}$ K $^{-1}$ at 10 T). Figure 9-(b) depicts the low doping regime ($x = 0.05$), where Ni^{3+} ions (green circles) substitute Mn^{3+} ions at the B-sites, introducing local lattice distortions and disrupting the Mn–O–Mn network. The red markers (X) indicate broken FM bonds, showing the disruption of the ferromagnetic double-exchange pathways. This substitution reduces the FM interactions while enhancing $\text{Ni}^{3+}\text{--O--Ni}^{3+}$ AFM superexchange, leading to a suppression of the magnetic order. Consequently, the transition

temperature decreases to 44 K, and the magnetocaloric effect is significantly reduced ($\Delta S_{\max} = 0.98 \text{ J kg}^{-1} \text{ K}^{-1}$ at 10 T). Figure 9- (c) illustrates the high doping regime ($x = 0.10$), where a higher concentration of Ni^{3+} ions leads to the emergence of $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ double-exchange interactions (indicated by blue arrows), which supplement the weakened $\text{Mn}^{3+}\text{--O--Mn}^{3+}$ ferromagnetic coupling and partially restore the FM interactions. However, $\text{Ni}^{3+}\text{--O--Ni}^{3+}$ AFM interactions also strengthen, creating a complex competition between FM and AFM interactions. This results in a partial recovery of the magnetic properties, with T_N increasing to 49 K and $\Delta S_{\max}$ recovering to $1.15 \text{ J kg}^{-1} \text{ K}^{-1}$ at 10 T. The schematic thus illustrates the non-monotonic evolution of magnetic interactions with Ni doping, from the well-ordered A-type AFM structure in the undoped system to the disrupted FM network at low doping, followed by partial recovery at higher doping through $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ double exchange.

The non-monotonic nature of magnetic characteristics in dependence on Ni doping can be explained as follows:

1. Low concentration of doping ($x=0.05$): substitution of Mn^{3+} by Ni^{3+} destroys the Mn-O-Mn spin structure. It leads to weakening of FM double exchange ($\text{Mn}^{3+}\text{--O--Mn}^{4+}$) and strengthening of AFM superexchange ($\text{Ni}^{3+}\text{--O--Ni}^{3+}$). Therefore, T_N decreases.
2. High concentration of doping ($x=0.10$): the $\text{Ni}^{3+}\text{--O--Mn}^{3+}$ double exchange becomes dominant. At the same time, the $\text{Ni}^{3+}\text{--O--Ni}^{3+}$ AFM interactions become strong as well, forming a complicated competition.

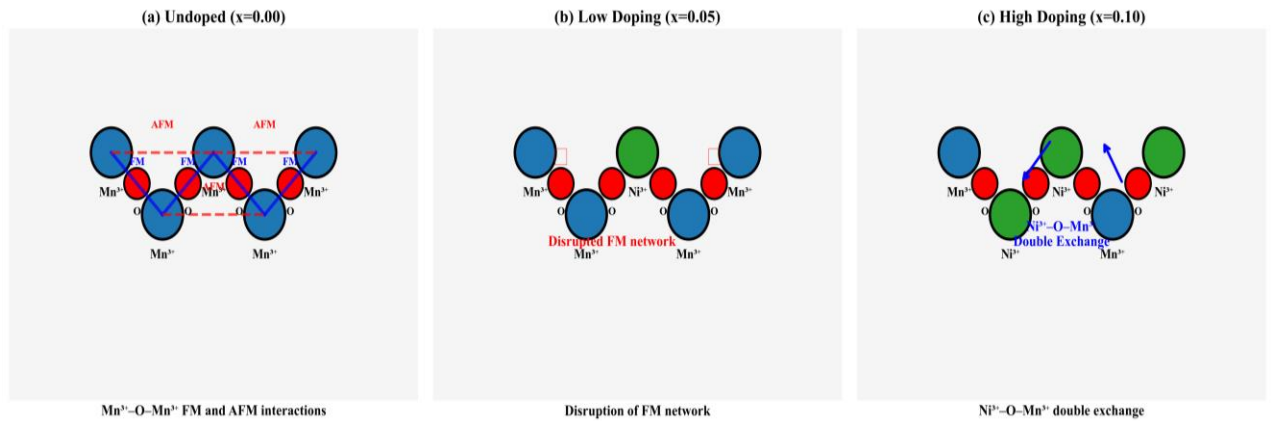


Figure 9: Schematic representation of magnetic interactions in $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$: (a) Undoped ($x=0.00$) showing $\text{Mn}^{3+}\text{-O-Mn}^{3+}$ FM and AFM interactions, (b) Low doping ($x=0.05$) showing disruption of FM network, (c) High doping ($x=0.10$) showing $\text{Ni}^{3+}\text{-O-Mn}^{3+}$ double exchange.

Conclusions

In the detailed Monte Carlo simulations performed on Ni-doped GdMnO_3 , we have been able to establish that the classical Heisenberg model using exchange parameters from the experimentally determined transition temperatures is indeed capable of accurately explaining the magnetic behavior of the system, with transition temperatures matching experimental results to within 0.5%. The magnetocaloric effect shows interesting possibilities for cryogenic technology applications, with maximum entropy changes at 10 T of $1.23 \text{ J kg}^{-1} \text{ K}^{-1}$ for the undoped compound ($x=0.00$), $0.98 \text{ J kg}^{-1} \text{ K}^{-1}$ for $x=0.05$, and $1.15 \text{ J kg}^{-1} \text{ K}^{-1}$ for $x=0.10$. Relative cooling power for these compounds at 10 T (45.7 , 38.2 , and 42.5 J kg^{-1} , respectively) further confirms that they can be used as materials in magnetic refrigeration, with the undoped composition having the greatest cooling capability. Values of the obtained critical exponents from the scaling analysis ($\beta \approx 0.325$ - 0.338 and $\gamma \approx 1.237$ - 1.310) coincide very well with those predicted by the 3D Ising universality class, which is an indication of the short-range magnetic interactions in these systems. Complex behavior of magnetic properties with respect to the increasing of the doping level is a result of the competition between different magnetic interactions (ferromagnetic double exchange ($\text{Mn}^{3+}\text{-O-Mn}^{3+}$, $\text{Mn}^{3+}\text{-O-Ni}^{3+}$), antiferromagnetic superexchange ($\text{Ni}^{3+}\text{-O-Ni}^{3+}$, $\text{Mn}^{3+}\text{-O-Mn}^{3+}$) and effects of Jahn-Teller distortions). While the most effective magnetocaloric effect is reached in the undoped system ($x=0.00$), the return to the observed parameters in the sample with $x=0.10$ indicates that even more effective cooling can be achieved through the usage of higher doping concentrations or some other strategy. Excellent coincidence of the results of Monte Carlo simulations and experimental data confirms our theoretical predictions and sets a reliable method for the prediction and optimization of the magnetocaloric properties of doped manganites.

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Conflict of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Chatgpt in order to improve language and readability with caution. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Declaration of interests

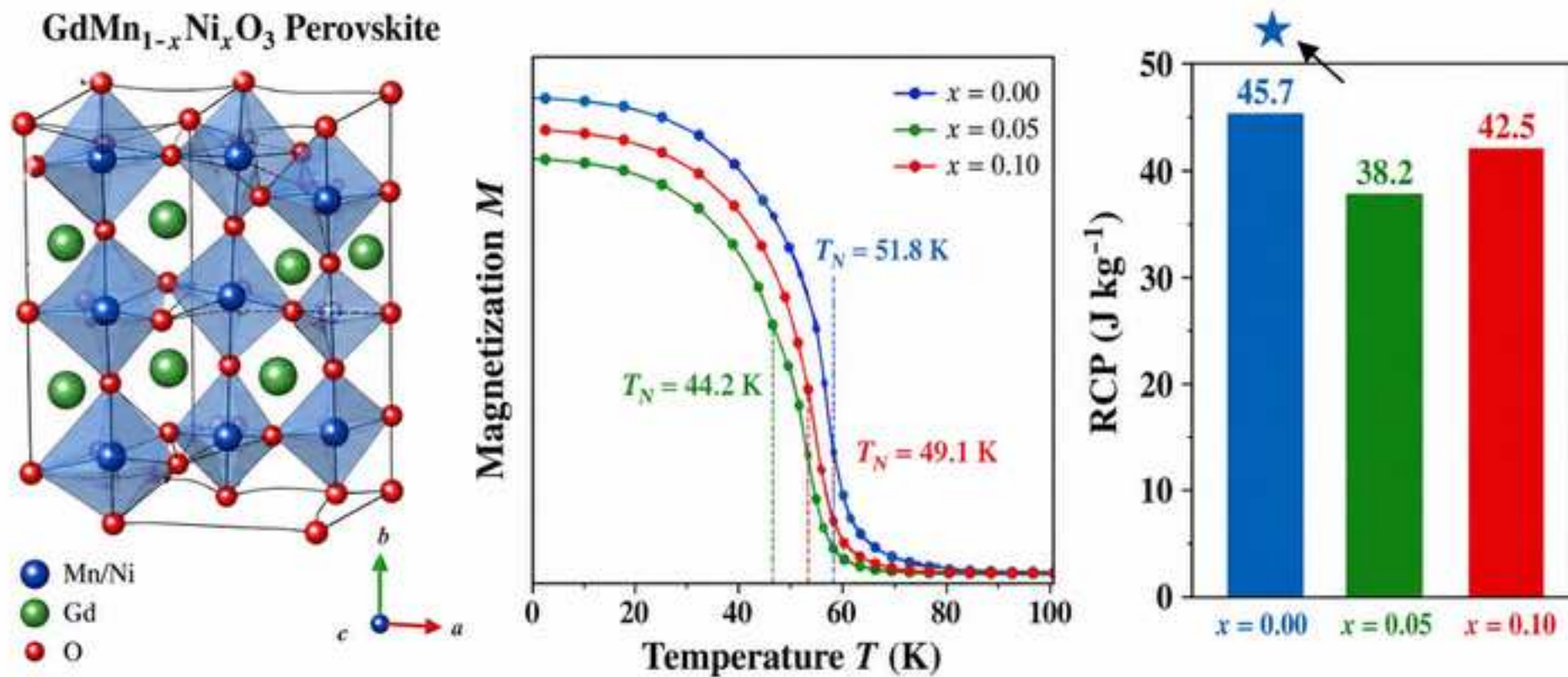
☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights:

- Classical Monte Carlo simulations reveal magnetism in Bi-doped GdMnO₃.
- Ni doping tunes Neels temperature and magnetocaloric performance.
- Simulated transition temperatures closely reproduce experimental results.
- Critical exponents confirm 3D Ising universality and short-range exchange.
- Exchange competition explains the non-monotonic magnetic behavior.

Monte Carlo Simulation of $\text{GdMn}_{1-x}\text{Ni}_x\text{O}_3$



3D Ising Universality Class | Cryogenic Magnetic Refrigeration (20-60 K)